

Section 2, Question 1

Real-Time Log Analytics as a MapReduce Batch Job

Distributed Systems — Homework 3

1 What the question asks

Homework 2 Question 7 computed a set of statistics over a server log in a single sequential program. This question asks for the same statistics, computed as a **MapReduce job** so that the work is divided across machines.

The input is N records, each a line of seven numbers:

```
timestamp  server_id  endpoint_id  user_id  status_code  response_time_ms
                                     bytes_sent
```

The required output is fixed, and includes total, successful and failed request counts; the average, minimum and maximum response time; total bytes; counts per HTTP status class; the busiest 60-second interval; and the top K servers and endpoints by request count. The answer must match the sequential program **byte for byte**, whatever the input is divided into.

2 The idea the solution rests on

Every quantity above is a **count**, a **sum**, a **minimum** or a **maximum**. All four share one property: they can be computed on two halves of the data independently and combined afterwards, in any order, giving the same answer. That property — mergeability — is what makes the problem expressible as MapReduce at all.

An average is *not* mergeable, so an average is never stored. The mapper keeps a sum and a count; the division happens once, in the reducer, when printing.

Response times are stored as **whole numbers of millionths of a millisecond** (12.5 ms becomes 12,500,000). Floating-point addition is not associative, and map tasks finish in a different order on every run, so decimals could produce a different final digit each time. Integers cannot. One of the edge-case inputs contains values as small as 10^{-6} ms, which sets how fine the unit has to be.

3 What was implemented

Three standalone C++ programs, each reading standard input and writing standard output — which is exactly Hadoop Streaming's contract, so the same binaries run under Hadoop unchanged.

Program	Role	Lines
mapper.cpp	reads a split, counts records, emits one line per distinct key	60
combiner.cpp	merges a mapper's own lines before they travel	20
reducer.cpp	adds all mappers' lines, computes averages and Top- K , prints	95
analytics.h/.cpp	the shared Stats record and the wire format	173

3.1 The pipeline

`split` → `mapper` (one per machine) → `sort` (per machine) → `combiner` (per machine) → `sort` (global)
→ `reducer` (one)

Four kinds of key travel between the stages: **G** for the global counters, **S:id** for one server, **E:id** for one endpoint, and **T:minute** for one 60-second window. The sort *is* the shuffle — no program performs it.

3.2 Two decisions worth defending

In-mapper combining. The mapper does not emit a line per record. It accumulates in memory and emits one line per *distinct key* at the end of its split. On the ten-million-record input this makes the data crossing the network $278\times$ smaller than the input (396 MB in, 1.43 MB out). Moving data between machines is the expensive part of MapReduce, so this is the single most consequential choice in the design.

Exactly one reducer. Top- K requires seeing every server’s count before it can name the largest ten. Two reducers would each produce their own top ten, and merging those is not the same answer. The single reducer is cheap: 0.07 s against a 16 s map stage.

4 How it was run

The assignment names Apache Hadoop with YARN. On the RCE cluster that is not available — the shared HDFS is read-only for a student account and no ResourceManager is configured — and the course confirmed the Hadoop environment is unavailable, directing students to a Slurm-based script instead.

The benchmark therefore follows the shape of the course’s own `Mapreduce_distributed.sh`: `split` the input, `srun` a mapper on each machine, sort locally, combine, sort globally, reduce once.

Measurements use **one map task per machine**, sweeping 1, 2, 4 and 6 machines. This matters: packing several tasks onto one machine makes them share a physical core, and the resulting curve describes the scheduler rather than the program. It is also the allocation shape Assignment 2 used for the MPI version of this workload, which is what allows the two to be compared.

5 Results

Ten million records, median of three runs, every run diffed against the sequential program.

Machines	Total	Map	Sort	Combine	Gather	Reduce	Speedup
1	16.940 s	16.351	0.218	0.173	0.141	0.069	1.00×
2	9.985 s	9.478	0.160	0.124	0.152	0.071	1.70×
4	5.231 s	4.760	0.122	0.111	0.164	0.073	3.24×
6	3.637 s	3.172	0.114	0.102	0.174	0.075	4.66×

The map stage is the job, and it is CPU-bound. Mapping accounts for 16.35 of 16.94 seconds — 97%. It would be easy to assume the cost is reading 396 MB from disk, but measuring separately on a compute node, `cat` reads the same file in 0.26 s while the mapper takes 13.26 s, of which 13.16 s is user CPU time. The work is parsing ten million lines of text, not fetching them. This is why the program scales with machines at all.

The serial tail is real but small. The global gather, sort and reduce do not shrink as mappers are added — they grow slightly, from 0.21s to 0.25s, because more mappers emit more partial rows for the same keys. That is 1.2% of the one-machine run but 6.9% of the six-machine run, and it is what bends the speedup curve away from linear. This is Amdahl’s law made visible.

Small inputs should not be distributed. At 100,000 records, six machines return only $1.21\times$ over one. Process startup and five pipeline stages cost more than the work being divided. MapReduce carries a fixed overhead that is amortised only from roughly a million records upward.

Q1 - MapReduce (HW2 Q7 analytics, C++ Hadoop Streaming)

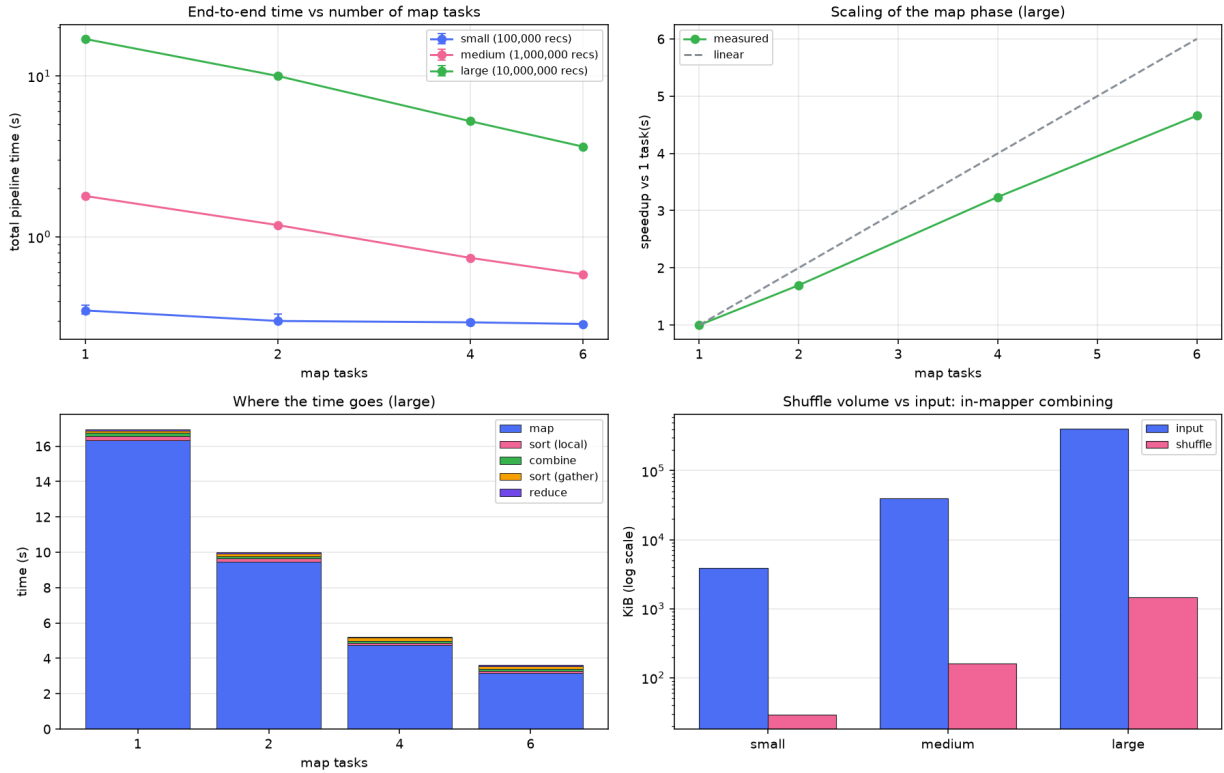


Figure 1: End-to-end time against machine count for each dataset; speedup against the linear ideal; where the time is spent, stacked by stage; and the volume of intermediate data against the input, showing the effect of in-mapper combining.

5.1 Comparison with MPI

The same analytics written with MPI, on the same machines, one process each.

Machines	MPI	MapReduce	MapReduce is
1	7.441 s	16.940 s	$2.28\times$ slower
2	4.356 s	9.985 s	$2.29\times$ slower
4	2.401 s	5.231 s	$2.18\times$ slower
6	1.701 s	3.637 s	$2.14\times$ slower

MPI is faster at every size, which is the expected outcome: it performs no shuffle, no sort, no serialisation to text and no process creation per stage. It holds state in memory and reduces once at the end.

Two things are more interesting than the headline. First, **the gap narrows** as machines are added, $2.28\times$ down to $2.14\times$. Second, MapReduce scales slightly *better* over this range — $4.66\times$

against MPI's $4.37\times$ — because its larger constant costs are precisely the part that parallelises, whereas MPI's closing reduction does not.

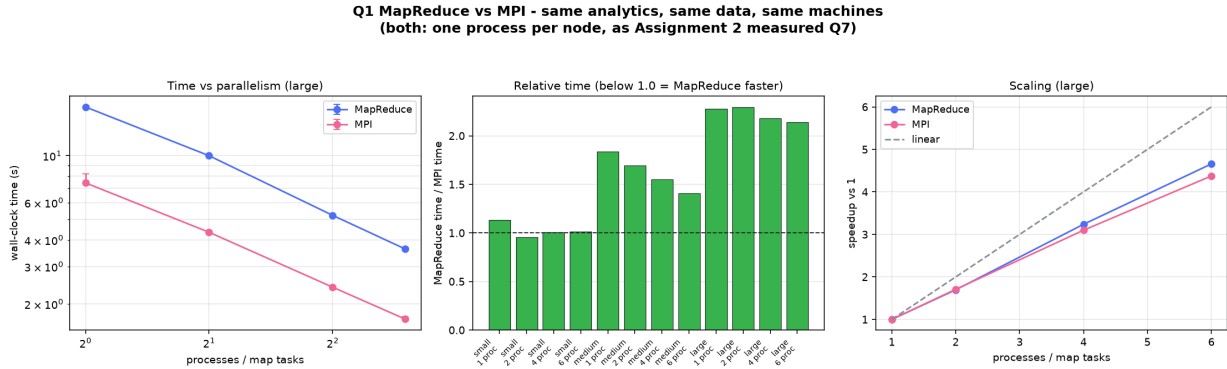


Figure 2: MapReduce against MPI on identical work and identical machines. Left: wall-clock time against machine count. Centre: the ratio, above 1.0 throughout but falling. Right: both scale comparably, so the difference lies in the constant factor rather than in the decomposition.

5.2 Where the constant factor comes from

The two programs differ in how they represent state, not in their parallel structure.

Program	State	Peak memory	Time
HW2 sequential	two passes, all records buffered, dense arrays	395 MB	4.65 s
MPI, one rank	the same, with a byte-range split	403 MB	6.31 s
Mapper (this work)	one streaming pass, sparse maps	10.1 MB	13.00 s

The Homework 2 design pre-scans the input to learn the range of server and endpoint identifiers, allocates dense arrays, and then indexes them directly: one array subscript per record, which is as fast as this computation can be. The price is holding all ten million records in memory first.

A mapper cannot do this. It sees only its own split and therefore cannot know the global identifier ranges in advance, so it counts in a single streaming pass into sparse associative containers, discarding each record once counted. The price is a tree lookup per record instead of an array subscript.

The result is a $39\times$ **reduction in memory for roughly $2.8\times$ the time**. This is not a defect in either program but a genuine engineering trade — and the memory property is exactly what permits the decomposition in the first place. A design needing 395 MB per task and a prior pass over the whole input cannot be split across mappers that each see a fraction of the data.

6 Correctness

Thirty-four checks, all passing, each diffed byte-for-byte against the unmodified Homework 2 sequential program kept in **reference/**. They cover the assignment's worked example, five edge cases (empty input, a single record, ties in Top- K , $K = 0$, and response times of 10^{-6} ms), and the same input divided 1, 2, 4 and 8 ways — because a correct decomposition must give an identical answer however the input is divided. All 36 runs of the performance sweep matched as well.

7 Conclusions

The workload suits MapReduce because every quantity it computes is mergeable, and the in-mapper combining that mergeability permits reduces network traffic by $278\times$. The implementation scales $4.66\times$ on six machines, limited by a small but growing serial tail rather than by the map stage.

It is about $2.2\times$ slower than an equivalent MPI program, and the reason is identifiable and defensible: sparse streaming state instead of dense buffered state, which costs time and saves $39\times$ the memory. That trade is not incidental to MapReduce — it is what makes a job splittable across machines that never see the whole input.